

◆ What is a computer?

- ❑ a computer is a sophisticated electronic calculating machine that:
 - ◆ **Accepts** input information,
 - ◆ **Processes** the information according to a list of internally stored instructions and
 - ◆ **Produces** the resulting output information.
- ❑ Functions performed by a computer are:
 - ◆ **Accepting** information to be processed as **input**.
 - ◆ **Storing** a list of **instructions** to process the information.
 - ◆ **Processing** the **information** according to the list of instructions.
 - ◆ **Providing** the results of the processing as **output**.
- ❑ What are the functional units of a computer?

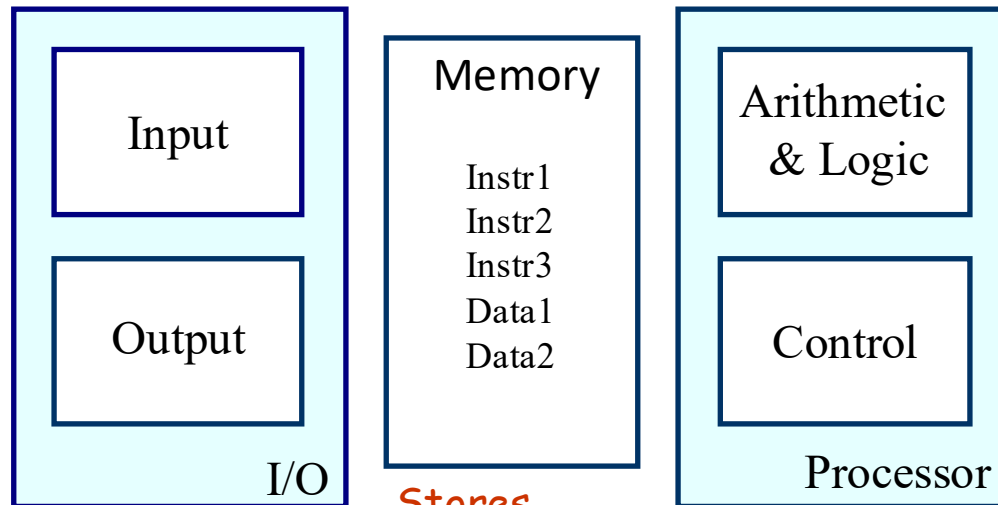
◆ Functional units of a computer

Input unit accepts information:

- Human operators,
- Electromechanical devices (keyboard)
- Other computers

Arithmetic and logic unit(ALU):

- Performs the desired operations on the input information as determined by instructions in the memory



Output unit sends results of processing:

- To a monitor display,
- To a printer

Stores information:

- Instructions,
- Data

Control unit coordinates various actions

- Input,
- Output
- Processing

◆ Information in a computer -- *Instructions*

- ❑ Instructions specify commands to:
 - ◆ **Transfer** information within a computer (e.g., from **memory** to **ALU**)
 - ◆ **Transfer** of information between the **computer** and **I/O** devices (e.g., from keyboard to computer, or computer to printer)
 - ◆ **Perform arithmetic** and **logic operations** (e.g., Add two numbers, Perform a logical AND).
- ❑ A sequence of instructions to perform a task is called a **program**, which is stored in the memory.
- ❑ **Processor fetches instructions** that make up a program from the memory and **performs** the **operations** stated in those instructions.
- ❑ What do the instructions operate upon?

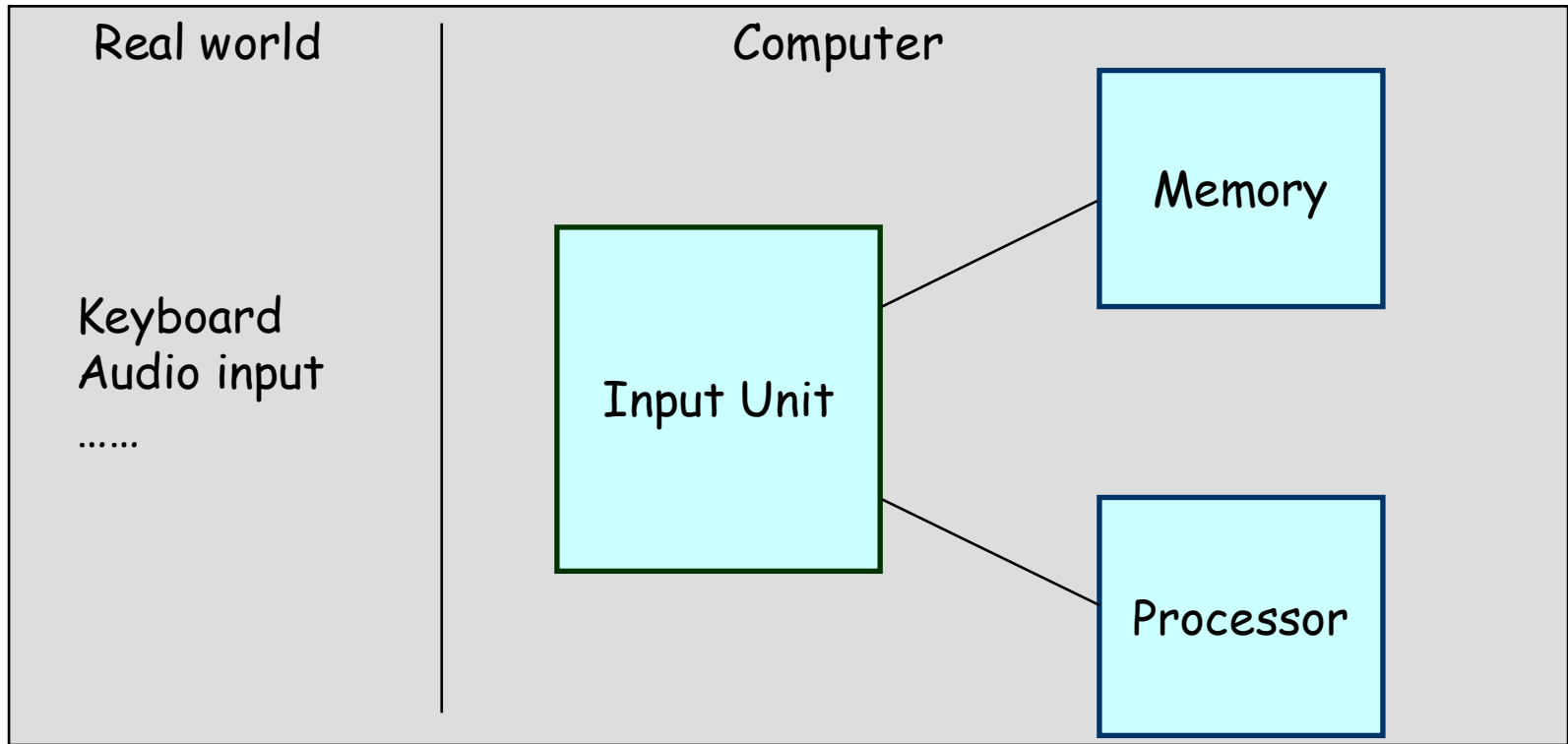
◆ Information in a computer -- Data

- ❑ Data are the “operands” upon which instructions operate.
- ❑ Data could be:
 - ◆ Numbers,
 - ◆ Encoded characters.
- ❑ Data, in a broad sense means any digital information.
- ❑ Computers use data that is encoded as a string of binary digits called bits.

◆ Input unit

Binary information must be presented to a computer in a specific format. This task is performed by the **input unit**:

- **Interfaces** with input devices.
- **Accepts** binary information from the input devices.
- **Presents** this binary information in a format expected by the computer.
- **Transfers** this information to the memory or processor.



◆ Memory unit

- ❑ Memory unit stores **instructions** and **data**.
 - ◆ Recall, data is represented as a series of bits.
 - ◆ To store data, memory unit thus stores **bits**.
- ❑ **Processor** reads **instructions** and reads/writes **data** from/to the **memory** during the **execution** of a program.
 - ◆ In theory, **instructions** and **data** could be fetched one bit at a time.
 - ◆ In practice, a **group of bits** is fetched at a time.
 - ◆ Group of bits stored or retrieved at a time is termed as "**word**".
 - ◆ Number of bits in a word is termed as the "**word length**" of a computer.
- ❑ In order to **read/write** to and from **memory**, a processor should know where to look:
 - ◆ "**Address**" is associated with each **word** location.

◆ Memory unit (contd..)

- ❑ Processor reads/writes to/from memory based on the memory address:
 - ◆ Access any word location in a short and fixed amount of time based on the address.
 - ◆ Random Access Memory (RAM) provides fixed access time independent of the location of the word.
 - ◆ Access time is known as "Memory Access Time".
- ❑ Memory and processor have to "communicate" with each other in order to read/write information.
 - ◆ In order to reduce "communication time", a small amount of RAM (known as Cache) is tightly coupled with the processor.
- ❑ Modern computers have three to four levels of RAM units with different speeds and sizes:
 - ◆ Fastest, smallest known as Cache
 - ◆ Slowest, largest known as Main memory.

◆ Memory unit (contd..)

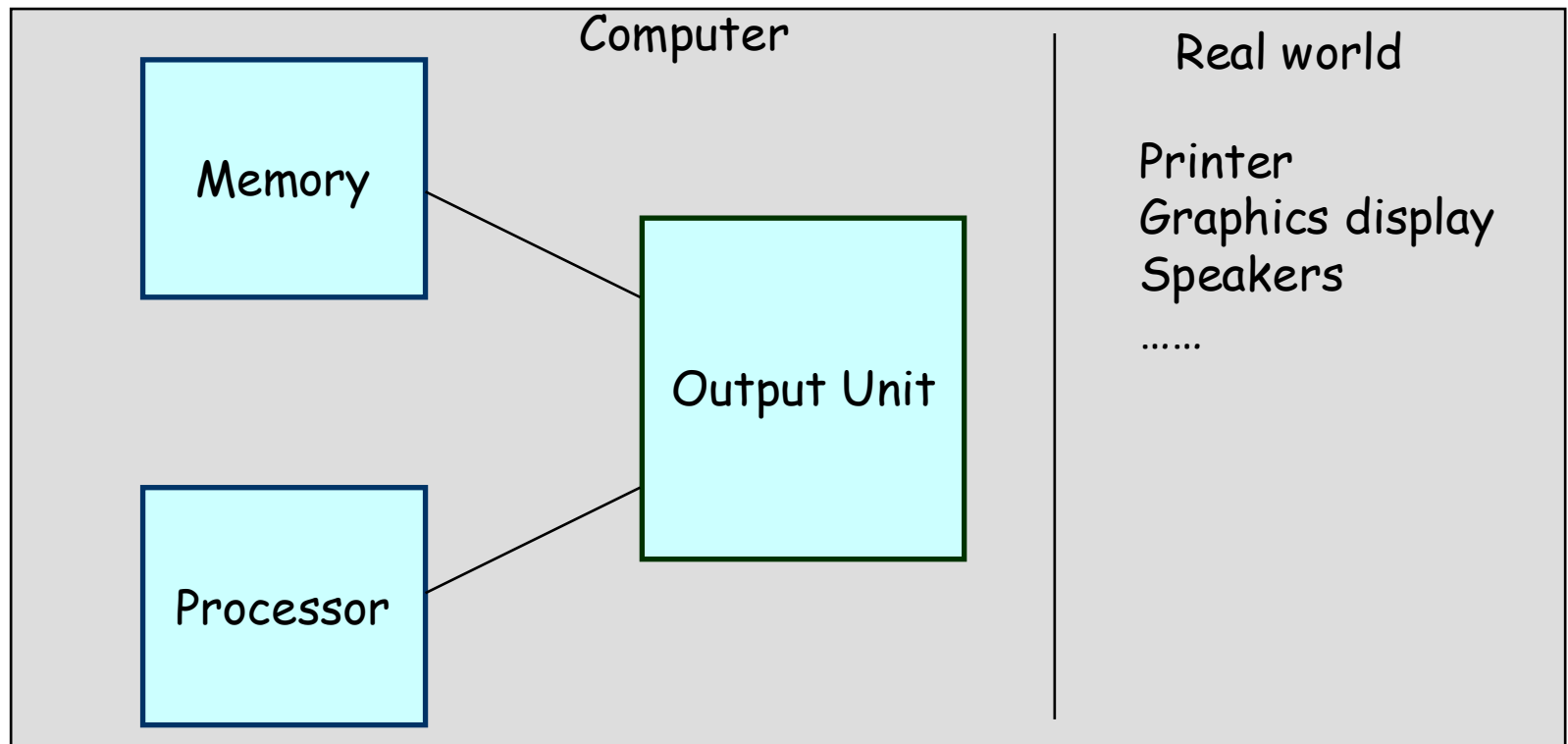
- ❑ Primary storage of the computer consists of RAM units.
 - ◆ Fastest, smallest unit is Cache.
 - ◆ Slowest, largest unit is Main Memory.
- ❑ Primary storage is insufficient to store large amounts of data and programs.
 - ◆ Primary storage can be added, but it is expensive.
- ❑ Store large amounts of data on secondary storage devices:
 - ◆ Magnetic disks and tapes,
 - ◆ Optical disks (CD-ROMS).
 - ◆ Access to the data stored in secondary storage is slower, but take advantage of the fact that some information may be accessed infrequently.
- ❑ Cost of a memory unit depends on its access time, lesser access time implies higher cost.

◆ Arithmetic and logic unit (ALU)

- ❑ Operations are executed in the Arithmetic and Logic Unit (ALU).
 - ◆ Arithmetic operations such as addition, subtraction.
 - ◆ Logic operations such as comparison of numbers.
- ❑ In order to execute an instruction, operands need to be brought into the ALU from the memory.
 - ◆ Operands are stored in general purpose registers available in the ALU.
 - ◆ Access times of general purpose registers are faster than the cache.
- ❑ Results of the operations are stored back in the memory or retained in the processor for immediate use.

◆ Output unit

- Computers represent information in a specific binary form. **Output units:**
 - **Interface** with output devices.
 - **Accept** processed **results** provided by the computer in specific **binary** form.
 - **Convert** the information in binary form to a **form understood** by an output device.

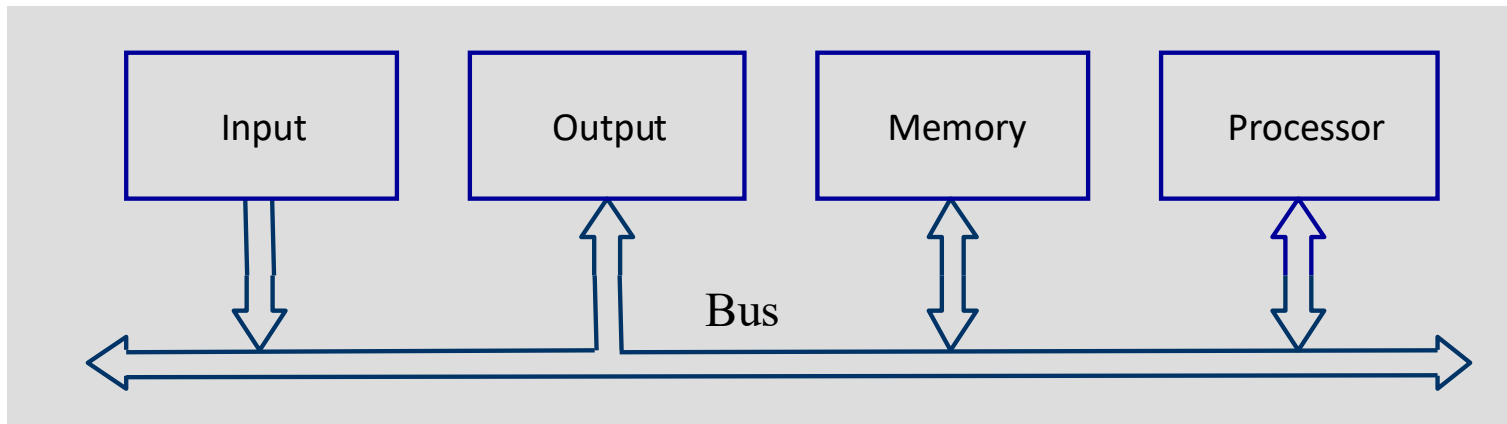


◆ Control unit

- ❑ Operation of a computer can be summarized as:
 - ◆ **Accepts** information from the input units (**Input** unit).
 - ◆ **Stores** the information (**Memory**).
 - ◆ **Processes** the information (**ALU**).
 - ◆ **Provides** processed results through the output units (**Output** unit).
- ❑ **Operations** of Input unit, Memory, ALU and Output unit are coordinated by **Control** unit.
- ❑ Instructions control "**what**" operations take place (e.g. data transfer, processing).
- ❑ **Control** unit generates **timing** signals which determines "**when**" a particular operation takes place.

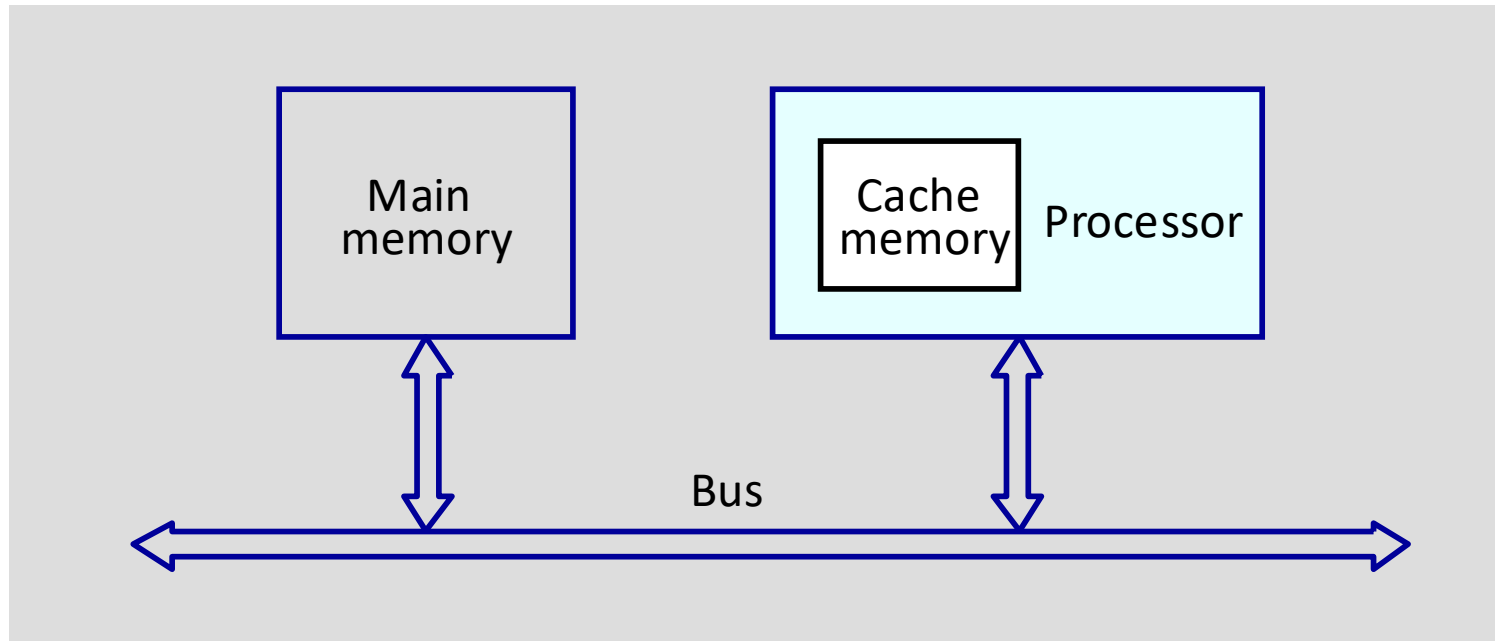
◆ How are the functional units connected?

- For a computer to achieve its operation, the **functional units** need to **communicate** with each other.
- In order to communicate, they need to be **connected**.



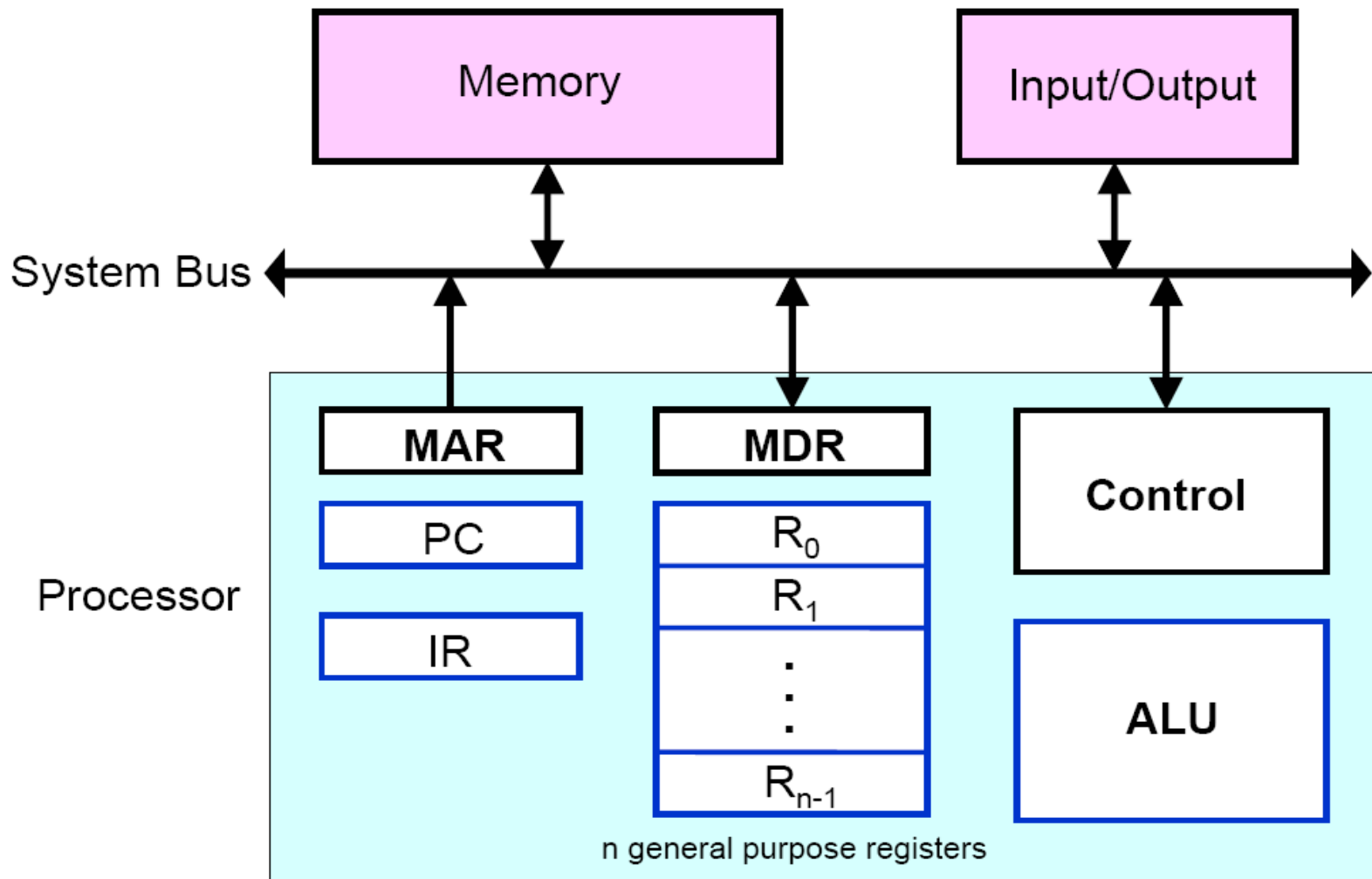
- Functional units may be connected by a **group of parallel wires**.
- The group of parallel wires is called a **bus**.
- Each **wire** in a bus can transfer **one bit** of information.
- The **number** of parallel **wires** in a bus is equal to the **word length** of a computer

♦ Organization of cache and main memory

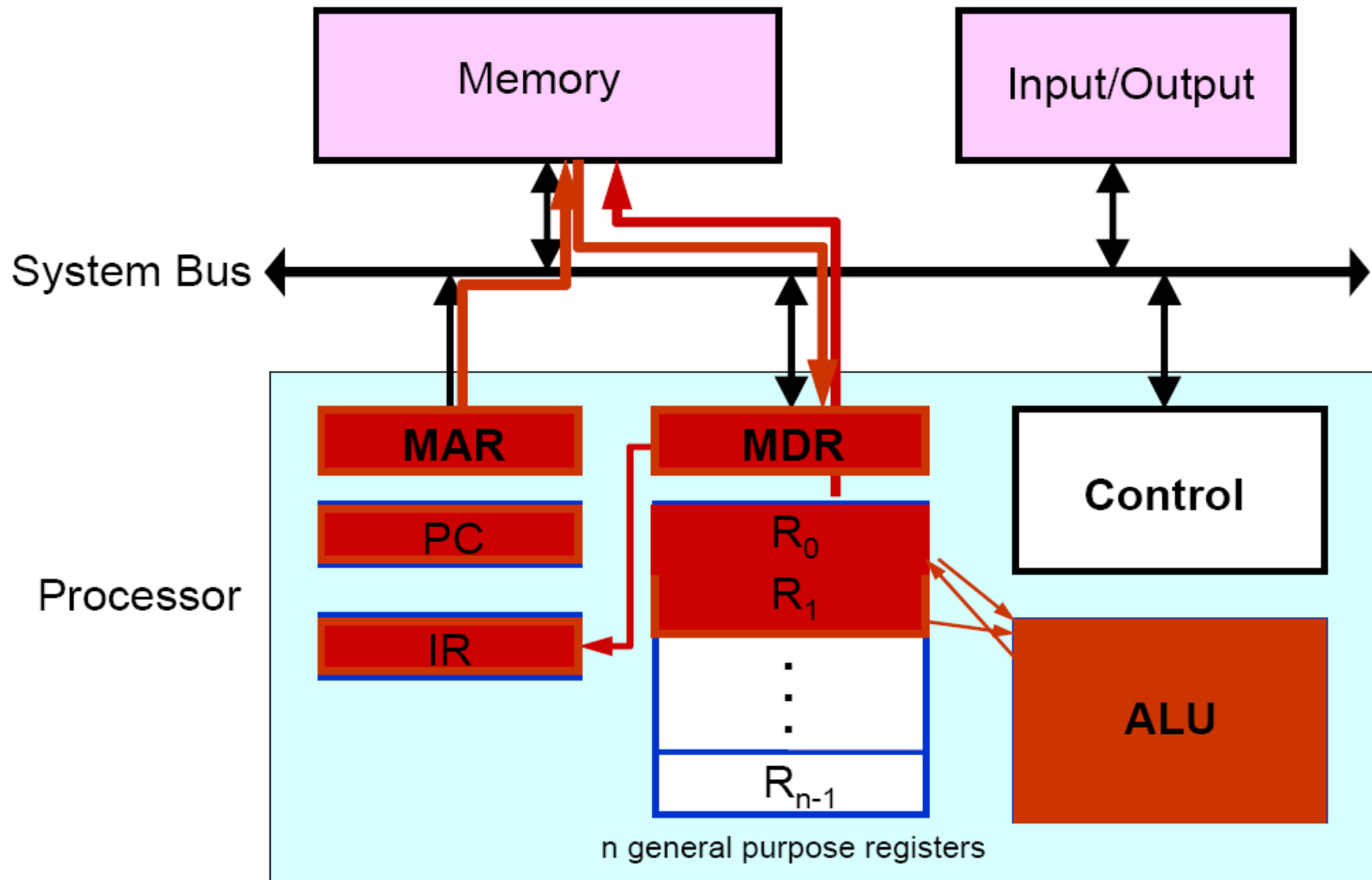


Why is the access time of the cache memory lesser than the access time of the main memory?

Computer Components: Top-Level View

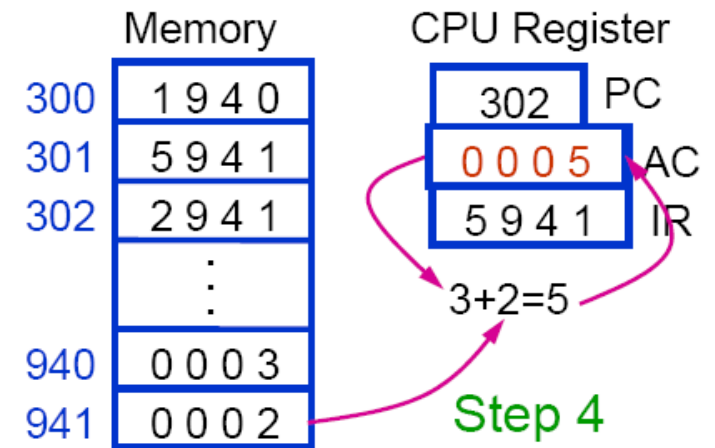
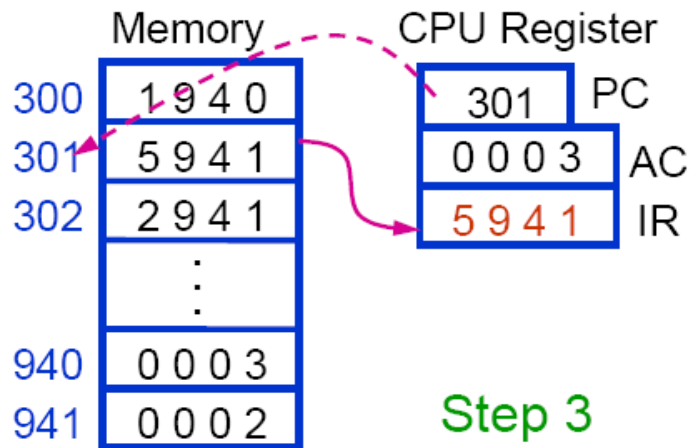
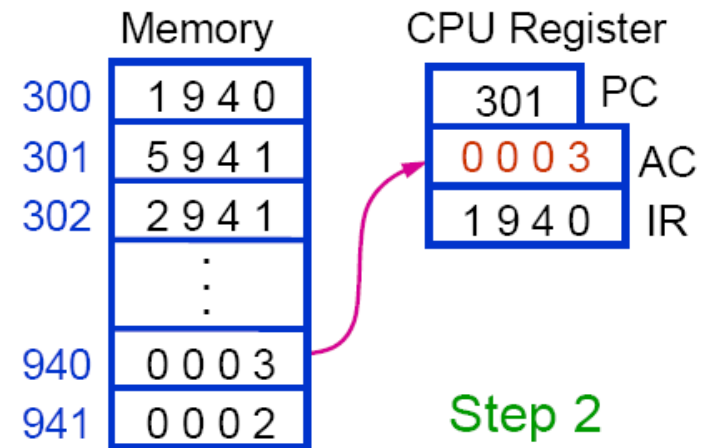
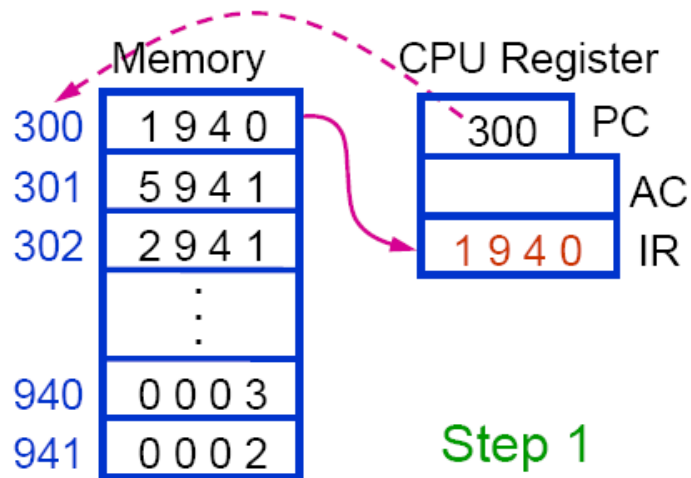


◆ Basic Operational Concepts

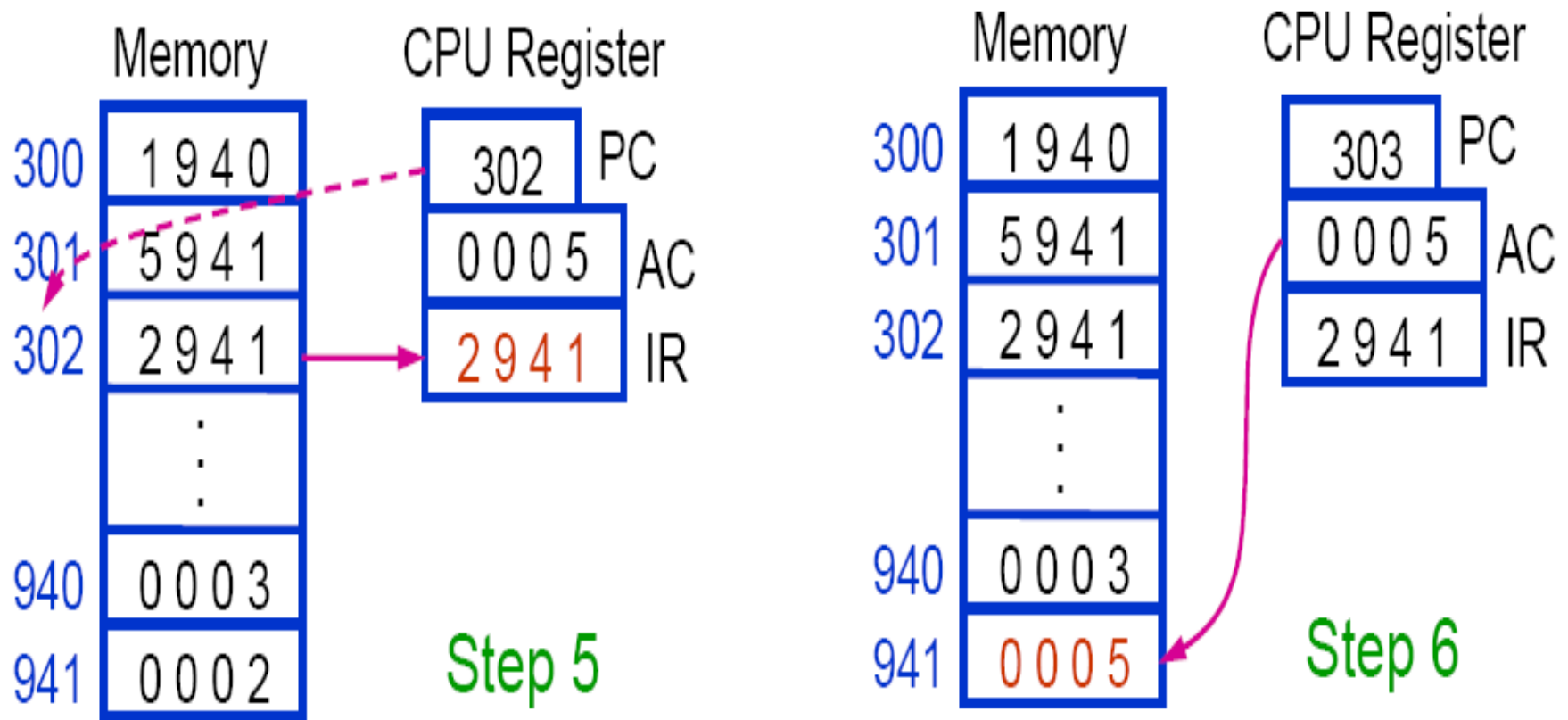




A Partial Program Execution Example



◆ A Partial Program Execution Example





Interrupt

- ❑ Normal execution of programs may be **interrupted** if some device requires **urgent** servicing
 - ◆ To deal with the situation immediately, the normal execution of the current program must be interrupted

 - ❑ **Procedure of interrupt** operation
 - ◆ The **device** raises an **interrupt signal**
 - ◆ The **processor** provides the requested service by **executing** an appropriate **interrupt-service routine**
 - ◆ The **state** of the **processor** is first **saved** before servicing the interrupt
 - Normally, the contents of the **PC**, the general **registers**, and some **control** information are stored in **memory**
 - ◆ When the interrupt-service routine is **completed**, the **state** of the **processor** is **restored** so that the interrupted program may continue
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◆ Classes of Interrupts

□ Program

- ◆ Generated by some condition that occurs as a result of an instruction execution such as arithmetic **overflow**, **division by zero**, attempt to execute an **illegal machine instruction**, or reference **outside** a user's allowed **memory** space

□ Timer

- ◆ Generated by a timer within the processor. This allows the operating system to **perform** certain **functions** on a **regular** basis

□ I/O

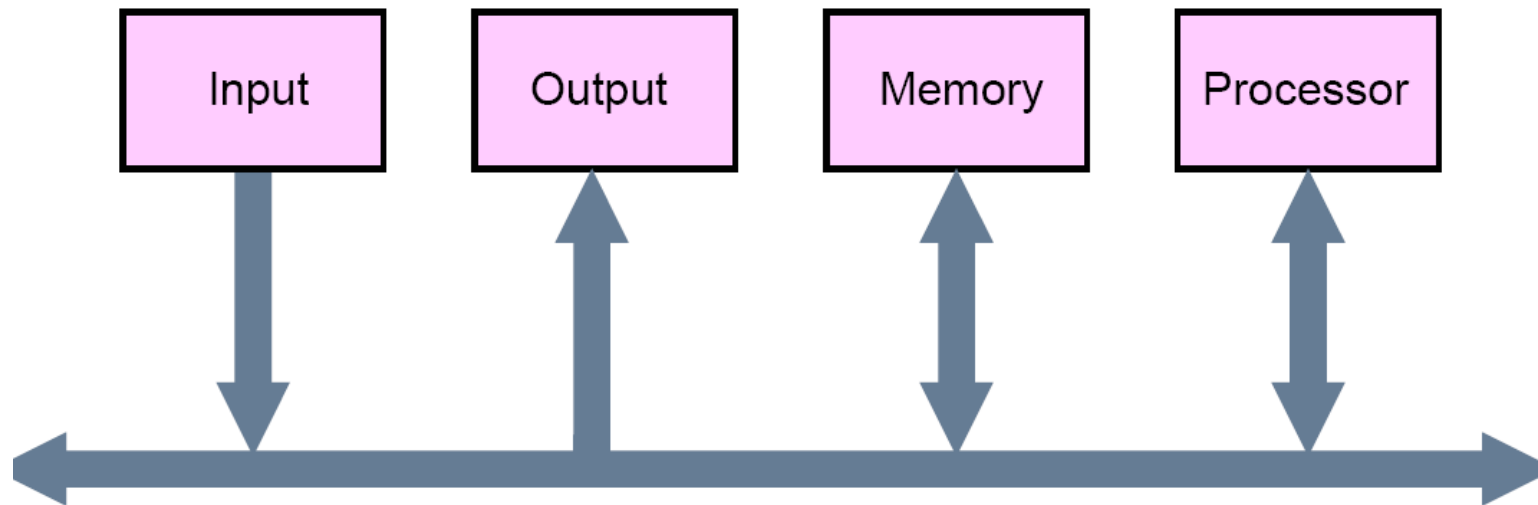
- ◆ Generated by an I/O controller, to **signal normal completion** of an operation or to **signal** a variety of **error conditions**

□ Hardware failure

- ◆ Generated by a failure such as **power failure** or **memory parity error**
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◆ Bus Structures

- A **group** of **lines** that serves a **connecting path** for several devices is called a **bus**
 - ◆ In addition to the **lines** that carry the **data**, the bus must have **lines** for **address** and **control** purposes
 - ◆ The simplest way to interconnect functional units is to use a **single bus**, as shown below





Drawbacks of the Single Bus Structure

- ❑ The **devices** connected to a bus **vary** widely in their **speed** of operation
 - ◆ Some devices are relatively **slow**, such as **printer** and **keyboard**
 - ◆ Some devices are considerably **fast**, such as **optical disks**
 - ◆ **Memory** and **processor** units operate are the **fastest** parts of a computer
 - ❑ Efficient transfer mechanism thus is needed to cope with this problem
 - ◆ A common **approach** is to include **buffer registers** with the devices to **hold** the **information** during **transfers**
 - ◆ An another **approach** is to use **two-bus** structure and an additional **transfer mechanism**
 - A **high-performance** bus, a **low-performance**, and a **bridge** for transferring the data between the two buses. ARMA Bus belongs to this structure
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Software

- ❑ In order for a user to enter and run an application program, the computer must already contain some **system software** in its **memory**

 - ❑ **System software** is a collection of **programs** that are executed as needed to **perform functions** such as
 - ◆ **Receiving and interpreting user commands**
 - ◆ **Running standard application programs** such as word processors, etc, or games
 - ◆ **Managing the storage and retrieval of files** in **secondary storage** devices
 - ◆ **Controlling I/O units** to receive input information and produce output results
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Software

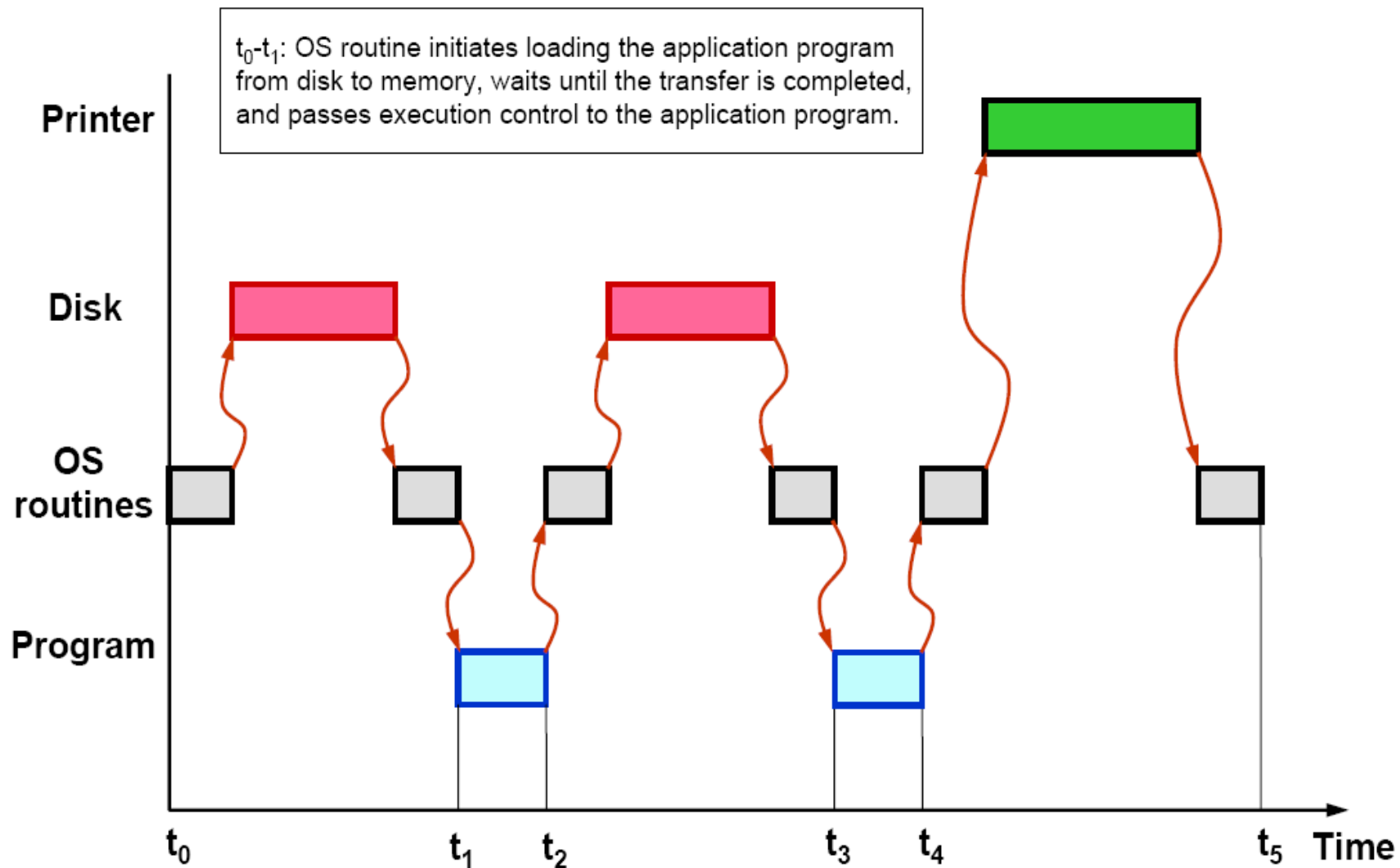
- ❑ Translating programs from **source** form prepared by the user into **object** form consisting of machine instructions
 - ❑ **Linking** and **running user-written** application **programs** with existing standard **library routines**, such as numerical computation packages
 - ❑ **System software** is thus responsible for the **coordination of all activities** in a computing system
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Operating System

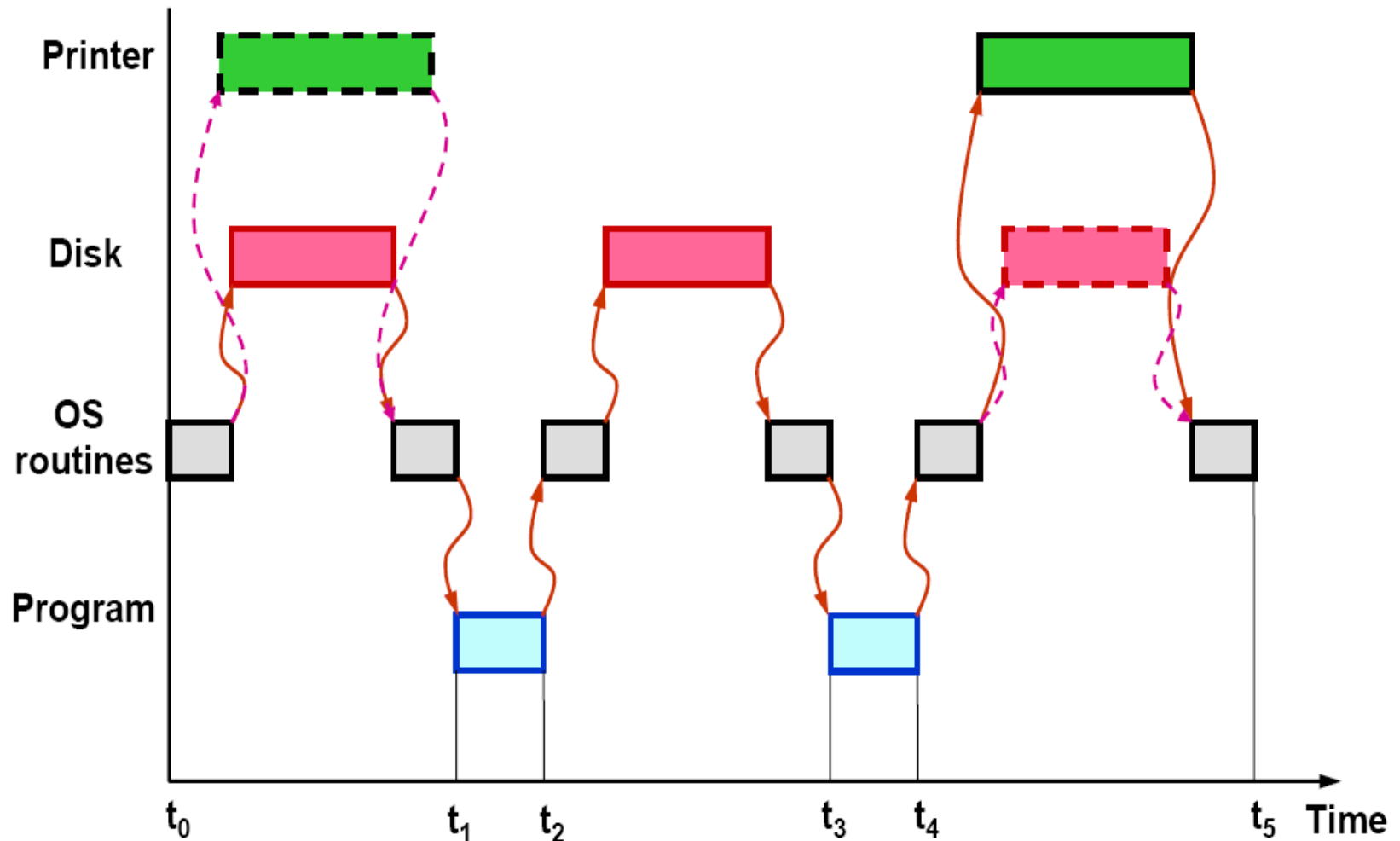
- ❑ Operating system (OS)
 - ◆ This is a large program, or actually a collection of routines, that is used to control the sharing of and interaction among various computer units as they perform application programs
 - ❑ The OS routines perform the tasks required to assign computer resource to individual application programs
 - ◆ These tasks include assigning memory and magnetic disk space to program and data files, moving data between memory and disk units, and handling I/O operations
 - ❑ In the following, a system with one processor, one disk, and one printer is given to explain the basics of OS
 - ◆ Assume that part of the program's task involves reading a data file from the disk into the memory, performing some computation on the data, and printing the results
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User Program and OS Routine Sharing



♦ Multiprogramming or Multitasking



◆ Performance

- ❑ The speed with which a computer executes programs is affected by the design of its hardware and its machine language instructions
 - ❑ Because programs are usually written in a high-level language, performance is also affected by the compiler that translates programs into machine languages
 - ❑ For best performance, the following factors must be considered
 - ◆ Compiler
 - ◆ Instruction set
 - ◆ Hardware design
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Performance

- ❑ Processor circuits are **controlled** by a **timing signal** called a **clock**
 - ◆ The clock defines regular time intervals, called clock cycles
 - ❑ To **execute** a machine **instruction**, the **processor** **divides** the **action** to be performed into a **sequence** of basic **steps**, such that **each step** can be completed in **one clock cycle**
 - ❑ Let the length P of one clock cycle, its inverse is the **clock rate**, $R=1/P$
 - ❑ Basic performance equation
 - ◆ $T=(N \times S)/R$, where T is the processor **time** required to execute a program, N is the **number** of **instruction** executions, and S is the average **number** of basic **steps** needed to execute one machine instruction
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Performance Improvement

- ❑ **Pipelining** and **superscalar** operation
 - ◆ **Pipelining**: by **overlapping** the **execution** of successive instructions
 - ◆ **Superscalar**: different instructions are **concurrently executed** with multiple instruction pipelines. This means that multiple functional units are needed
 - ❑ **Clock rate** improvement
 - ❑ Improving the **integrated-circuit technology** makes logic circuits **faster**, which **reduces** the **time** needed to complete a basic **step**
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◆ Performance Improvement

- ❑ Reducing amount of processing done in one basic step also makes it possible to **reduce** the **clock period**, P .
 - ❑ However, if the actions that have to be performed by an instruction remain the same, the number of basic steps needed may increase
 - ❑ **Reduce** the **number** of basic **steps** to execute
 - ◆ Reduced instruction set computers (RISC) and complex instruction set computers (CISC)
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